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United States Patent Application Publication

20250278994

Kind Code

A1

Publication Date

September 04, 2025

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SPACE-SAVING EQUIPMENT PANEL STATUS INDICATOR

Abstract

Aspects of the subject disclosure may include, for example, a process for providing status indication, that includes receiving a control signal at a proximal end of an elongated flexible circuit assembly including a distal portion attached to and extending along at least a portion of an elongated, panel-facing edge of a planar circuit board. The process further includes directing the control signal around a bent portion of the elongated flexible circuit assembly and to a number of visual status indicators distributed along at least a distal end of the elongated flexible circuit assembly, wherein at least a portion of the number of visual status indicators is actuated, responsive to the control signal, to provide a visual indication of a status of equipment supported by the planar circuit board. Other embodiments are disclosed.

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Appl. No.: 19/211730

Filed: May 19, 2025

Foreign Application Priority Data

IN 202311016582

Mar. 16, 2023

Related U.S. Application Data

parent US continuation 18320611 20230519 PENDING child US 19211730

Publication Classification

Int. Cl.: G08B21/18 (20060101); H05B45/20 (20200101); H05B45/30 (20200101)

U.S. Cl.:

CPC G08B21/18 (20130101); H05B45/20 (20200101); H05B45/30 (20200101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/320,611 filed May 19, 2023 by Gupta et al., entitled “SPACE-SAVING EQUIPMENT PANEL STATUS INDICATOR”, which claims priority under 35 U.S.C. § 1.119 to Indian patent application no. 202311016582, filed on Mar. 16, 2023. All sections of the aforementioned application(s) and/or patent(s) are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

[0002] The subject disclosure relates to a space-saving equipment panel status indicator.

BACKGROUND

[0003] Modern computing systems often include various network devices to facilitate network-based communications. For example, network devices such as switches and routers may be interconnected with each other and various networks to support such communications.

[0004] Individual network devices may include display panels to provide status information that may be monitored by a user. Frequently, these display panels are implemented with light emitting diodes (LEDs) configured to display the status of particular parameters of the device or the network system. For example, a first set of LEDs may be provided to indicate the operational status of various ports of the network device, and a second set of LEDs may be provided to indicate various configuration and/or system-related information. Frequently, such network devices are implemented in compact hardware enclosures having relatively small form factors that can facilitate convenient grouping of multiple devices, e.g., in equipment racks and/or cabinets of a data center.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0006] FIG. 1A is an elevation view illustrating a front panel of an example electronic device equipped with space-saving visual indicators in accordance with various aspects described herein.

[0007] FIG. 1B is a front perspective view of the example electronic device illustrated in FIG. 1A in accordance with various aspects described herein.

[0008] FIG. 1C is a section view illustrating a front portion of the example electronic device illustrated in FIG. 1A in accordance with various aspects described herein.

[0009] FIG. 2A is a perspective view illustrating a panel-facing surface of an example flexible status indicator assembly in accordance with various aspects described herein.

[0010] FIG. 2B is a section view of the example flexible status indicator assembly illustrated in FIG. 2A in accordance with various aspects described herein.

[0011] FIG. 2C is an exploded plan view illustrating a top edge of the example flexible status indicator assembly illustrated in FIGS. 2A-2B, in proximity to a portion of a main printed circuit board in accordance with various aspects described herein.

[0012] FIG. 3 is a more detailed elevation view illustrating the front panel of another example electronic device in accordance with various aspects described herein.

[0013] FIG. 4A is an elevation view illustrating a front panel the example electronic device illustrated in FIG. 3, equipped with space-saving visual indicators in accordance with various aspects described herein.

[0014] FIG. 4B is a perspective view illustrating an interior portion of the example electronic device illustrated in FIG. 4A with the front panel removed in accordance with various aspects described herein.

[0015] FIG. 5 is a more detailed, exploded perspective view illustrating an interior portion of the example electronic device illustrated in FIG. 3 and FIGS. 4A-4B in accordance with various aspects described herein.

[0016] FIG. 6A is an elevation view illustrating a front panel of another example electronic device equipped with space-saving visual indicators in accordance with various aspects described herein.

[0017] FIG. 6B is a section view illustrating a front portion of the example electronic device illustrated in FIG. 6A in accordance with various aspects described herein.

[0018] FIG. 7A is an elevation view illustrating a front panel of, yet another example electronic device equipped with space-saving visual indicators in accordance with various aspects described herein.

[0019] FIG. 7B is a section view illustrating a front portion of the example electronic device illustrated in FIG. 7A in accordance with various aspects described herein.

[0020] FIG. 8 is schematic diagram of an example status indicator system incorporating the example status indicator assembly of FIGS. 2A-2C in accordance with various aspects described herein.

[0021] FIG. 9 is timing diagram of an example control signal of the example status indicator circuit of FIG. 8 in accordance with various aspects described herein.

[0022] FIG. 10 depicts an illustrative embodiment of an equipment status indication process in accordance with various aspects described herein.

[0023] FIG. 11 depicts an illustrative embodiment of an equipment status indication control process in accordance with various aspects described herein.

[0024] FIG. 12 is a block diagram of an example, non-limiting embodiment of a computing environment in accordance with various aspects described herein.

DETAILED DESCRIPTION

[0025] The subject disclosure describes, among other things, illustrative embodiments of repeating, conductive patterns configured for coupling to a thermal source to transfer heat away from the thermal source according to a desired directionality. Repeating conductive patterns are thermally coupled to each other to combine the desired directionality of each of the patterns to obtain a distributed directionality transferring heat away from the thermal source to reduce any unwanted localized concentration of the heat. Other embodiments are described in the subject disclosure.

[0026] One or more aspects of the subject disclosure include an equipment panel indicator assembly. The equipment panel indicator assembly includes an elongated circuit board extending between a proximal end and a distal end, wherein the elongated circuit board includes an electrical interface at the proximal end. The elongated circuit board further includes an outer surface having length and width dimensions, such that the length dimension that is substantially greater than the width dimension. The indicator assembly further includes a number of electrically actuated indicators arranged along the outer surface and distributed along at least a proximal portion of the length dimension. The indicator assembly further includes a number of conductive traces extending between the electrical interface and at least a portion of the number of electrically

actuated indicators. The indicator assembly further includes a fastening enhancement feature, mechanism and/or device positioned along at least a portion of an inner surface of the elongated circuit board, wherein the fastening enhancement feature is configured for attachment to an edge of a planar structure, such as a printed circuit board.

[0027] One or more aspects of the subject disclosure include a process for providing status indication. The process includes receiving a control signal at a proximal end of an elongated flexible circuit assembly including a distal portion attached to and extending along at least a portion of an elongated, panel-facing edge of a planar circuit board. The process further includes directing the control signal around a bent portion of the elongated flexible circuit assembly and to a number of visual status indicators distributed along at least a distal end of the elongated flexible circuit assembly. At least a portion of the number of visual status indicators are actuated, responsive to the control signal, to provide a visual indication of a status of equipment supported by the planar circuit board.

[0028] One or more aspects of the subject disclosure include an electronic system, having a housing that includes an equipment panel observable by a human operator during operation. The electronic system further includes a primary circuit board including a number of circuit devices attached thereon, wherein the primary circuit board is disposed within the housing, having an elongated edge extending along an interior surface of the equipment panel. The electronic system further includes an elongated circuit assembly having an array of light emitting diodes extending along at least a distal portion of the elongated circuit assembly, wherein the elongated circuit assembly is attached to the primary circuit board, along the elongated edge, such that the elongated circuit assembly is located, positioned, secured, encapsulated and/or otherwise confined between the elongated edge and the interior surface of the equipment panel. The electronic system further includes a controller circuit configured to provide a status indicator control signal, wherein the controller circuit is in communication with the array of light emitting diodes via electrical traces of the elongated circuit assembly, the light emitting diodes actuated according to the status indicator control signal to provide a visual indication of a status of the number of circuit devices that is observable at the equipment panel.

[0029] Electronic devices, such as computing and/or networking devices may be configured with electronic circuitry. For example, a rectangular enclosure may include one or more printed circuit boards (PCBs) housing electronic circuits supporting functionality of the device. Such enclosures typically provide a relatively small front panel area where status indicators, e.g., LED display panels, may be located to facilitate convenient inspection of the status of various parameters indicated on the device. Quite often, enclosed functional circuits of these devices generate a substantial amount of heat driven by trends that incorporate greater processing power and operational speeds into the same small enclosures. In such instances, equipment enclosures typically incorporate one or more cooling features that facilitate management of thermal loads. Examples include ventilation panels that may be used alone or in combination with forced air cooling, e.g., a fan.

[0030] In many instances, one or more panels of an equipment enclosure incorporate openings, screens and/or vents to facilitate heat transfer. As panel space may be limited, that space dedicated to cooling features may be maximized. In at least some embodiments, virtually all available panel may be allocated to cooling features. It is understood that some panel space may be allocated to and/or otherwise reserved for non-cooling features, such as cable interconnections, e.g., network ports, operator controls and the like. Any remaining panel space would represent that are to be optimized and/or maximized for cooling. It may be further appreciated that, as networked computing systems increase in complexity, individual network devices may be required to display status information pertaining to increasing numbers of parameters, e.g., corresponding to increased numbers of ports or system information. The various examples provided herein provide features that present substantial numbers of status indicators along one or more panels of an equipment

enclosure, while preserving panel space for other applications, such as interconnects, operator controls and/or cooling.

[0031] FIG. 1A is an elevation view illustrating an example of a front panel **102** of a compact equipment enclosure of an example of an electronic device **100** equipped with space-saving visual indicators in accordance with various aspects described herein. Without limitation, the electronic device **100** can include any combination of electronic circuitry and/or electromechanical devices, generally configured to implement one or more functionalities. The front panel **102** may be viewable by a human observer and, as such, present a suitable location for presenting observable information as may vary from time to time either directly or remotely, e.g., via a video monitor. By way of example, the front panel **102** may present visual and/or audible indications that provide configuration and/or status information of the electronic device **100** and/or one or more sub-assemblies and/or components of the electronic device **100**. For example, the electronic device **100** may be configured for a telecommunication application, such as a router, a switch, a gateway and the like. It is not uncommon for such devices to support multiple redundant functionalities, such as multiple router ports and/or switch ports and/or network interfaces. It is envisioned that in at least some embodiments, the status information may identify configuration and/or status of the individual ports and/or interfaces, e.g., signifying whether a port is active, idle, transmitting, receiving and/or experiencing an error state.

[0032] According to the illustrative example, the front panel **102** is rectangular, although other shapes and/or sizes are contemplated. The front panel **102** extends over a surface area $A_{sub.Total}$, defined by the panel length $L_{sub.1}$ and height $H_{sub.1}$, i.e., with the total area determined according to the product of the length and height, i.e., $A_{sub.Total} = L_{sub.1} \times H_{sub.1}$. The front panel **102** also includes an example status indicator arrangement **108**. The status indicator arrangement **108** may include one or more visual status indicators **112**. The individual status indicators may include visual status indicators, audible status indicators or any combination of visual and audible status indicators. By way of example and without limitation, visual status indicators **112** may include one or more of an incandescent bulb, fluorescent lamps, neon lamps, light emitting diodes (LEDs), display panels, e.g., configured to display an image and/or video, touch panels, an alphanumeric indicator, and/or any combination thereof. By way of further example, audio indicators may include one or more of a buzzer, a speaker, a piezoelectric transducer, a bell, and the like.

[0033] In at least some embodiments, the front panel **102** may include one or more reserved areas. Any reserved area generally indicates that the area is unavailable to accommodate status indicators. In at least some embodiments, an area may be reserved according to a particular function, e.g., accommodating a physical network interface connector or port and/or ventilation features. For example, according to the illustrative embodiment, an upper portion of the front panel **102** includes a first group of reserved areas **104a**, **104b**, **104c**, generally **104**, while a lower portion of the front panel **102** includes a second group of reserved areas **106a**, **106b**, **106c**, generally **106**. Accordingly, these reserved areas **104**, **106** would be unavailable to accommodate any of the status visual indicators **112** of the status indicator arrangement **108**.

[0034] By way of example, one or more of the reserved areas **104**, **106** may be unavailable because it is occupied by another device, such as an electrical connector and/or an electromechanical port that would preclude positioning of a visual status indicator **112** there. Alternatively, or in addition, one or more of the reserved areas **104**, **106** may be unavailable because it is occupied by a vent, e.g., a ventilation panel as may be used for intake and/or exhaust air flow. In at least some embodiments, one or more of the reserved areas **104**, **106** may be unavailable because it is occupied by a design element, such as a label and/or branding mark. According to the foregoing examples, unavailability is due to the existence of a structure that may be observable from an observer of the front panel **102**.

[0035] It is conceivable that in at least some embodiments, unavailability may be due to a condition

occurring on an interior portion of the electronic device **100**, e.g., behind the front panel **102**. Such conditions may include, without limitation, a location of an electrical, mechanical and/or electromechanical assembly, device and/or component within an interior region of the electronic device **100** and proximate to an interior surface of the front panel **102**. The condition may arise from a mechanical interference, e.g., insufficient room to accommodate the visual status indicator **112**, and/or some other reason, such as a possibility of thermal interference and/or electrical interference, e.g., electromagnetic interference. It is understood that in such instances, the internal condition may not be visible by an observer of the front panel **102** but may nevertheless preclude positioning of a visual status indicator **112** there.

[0036] Each reserved area **104**, **106** represents a respective surface area $A_{sub.i}$, defined by a reserved area length $L_{sub.2}$ and height $H_{sub.2}$, i.e., the respective surface area $A_{sub.i} = L_{sub.1i} \times H_{sub.1i}$. A total reserved area $A_{sub.Reserved}$ may be determined as a sum of the individual reserved areas $A_{sub.i}$, i.e., $A_{sub.Reserved} = \sum A_{sub.i}$. In at least some embodiments, a ratio of reserved area to available area may be determined as a value $A_{sub.Reserved} / A_{sub.Total}$. Likewise, an unreserved area may be determined as a difference between the total area and the reserved area, i.e., $A_{sub.Unreserved} = A_{sub.Total} - A_{sub.Reserved}$. It is understood in such arrangements that the visual status indicators **112** would necessarily be positioned within the unreserved area of the front panel.

[0037] In at least some embodiments, a ratio of the reserved to unreserved areas may be maximized. For example, a maximum reserved area may be determined to accommodate a largest number of interconnections and/or communication ports as possible for a given surface area of the front panel **102**. Alternatively, or in addition, the maximum reserved area may be determined to accommodate the largest number of cooling vents possible. Such a maximization of reserved area may be determined according to the total surface area of the front panel **102**, less any portions thereof that may be excluded and/or otherwise unavailable to accommodate the communication ports and/or cooling vents.

[0038] In at least some embodiments, the front panel **102** may include one or more exclusion regions that may be unavailable for reservation according to the example reserved areas **104**, **106**. By way of example, such exclusion regions may include at least a portion of a peripheral region of the front panel **102** as may be occupied by structural supports, e.g., for attachment to top, bottom and/or side panels, and/or internal structures such as mounting frames and/or printed circuit boards. According to the illustrative example, the front panel **102** includes a bottom exclusion region **110** of a height h extending between a bottom perimeter of the front panel **102** and a bottom edge of the lower reserved areas **106**. Likewise, the example front panel **102** may include an upper exclusion region of the same or different height h extending between a top perimeter of the front panel **102** and an upper edge of the upper reserved areas **104**. Similarly, the example front panel **102** may include lateral exclusion regions, e.g., a left and/or right exclusion region having the same or different widths w extending between respective side perimeters of the front panel **102** and an adjacent side edges of the upper and lower reserved areas **104**, **106**.

[0039] According to the illustrative example, a central exclusion region **114** is provided along an elongated central region of the front panel **102**. The central exclusion region **114** occupies a height $H_{sub.4}$ and extends for a length $L_{sub.1}$, representing a central exclusion region area $A_{sub.Exclusion} = H_{sub.4} \times L_{sub.1}$. The central exclusion region **114** may be unavailable for reservation due to the location of an adjacent edge of a printed circuit board (PCB), e.g., a main PCB or motherboard. It is understood that in at least some embodiments, one or more of any identified exclusion regions, e.g., the central exclusion region **114**, may be available to accommodate one or more of the visual status indicators **112**, despite being unavailable to accommodate a reserved area. Indeed, at least one strategy that supports maximizing reserved area(s) **104**, **106**, while also providing visual status indicators **112**, takes advantage of locating at least a portion of the status indicators within the exclusion regions.

[0040] By way of example, the electronic circuitry and/or electromechanical devices of the example electronic device **100** may be housed at least partially within a supporting structure. Examples supporting structures may include, without limitation, a chassis, an open frame, a partially enclosed frame, and/or an enclosure, e.g., a box. Referring next to FIG. **1B**, a front perspective view of an example of an equipment enclosure **140** of an example device in accordance with various aspects described herein. The equipment enclosure **140** has six panels, including a front panel **142**, a top panel **144** and a right-side panel **146** as shown. The front panel corresponds to the example front panel **102** of the electronic device **100** (FIG. **1A**) Other panels include a left panel, a rear panel and a bottom panel, none of which are visible in the example illustration.

[0041] In at least some embodiments, the equipment enclosure **140** may be configured for mounting within another support structure, such as a frame and/or a cabinet. Examples include, without limitation, standard **19**-inch equipment racks adapted to secure one or more electronic devices in a stacked arrangement, e.g., a vertically stacked arrangement. It is common for equipment to accommodate such a standard form factor by incorporating a frame and/or enclosure configured to fit within and/or attach to such standard equipment racks. According to the illustrative example, the front panel **142** includes one or more mounting brackets, such as the example mounting flanges **148a**, **148b**, generally **148**. The flanges, in turn, may include one or more attachment and/or fastening enhancement features, mechanisms and/or devices, such as captive hardware, e.g., screws, slots and/or holes sized and positioned to accept mounting hardware, e.g., screws. When the equipment enclosure **140** is mounted within an equipment rack, one or more panels may be accessible, while other panels may be inaccessible. For example, a front and/or rear panel may be accessible for cable interconnects and/or human observation.

[0042] Although reference is made to a front panel, e.g., a front panel of a compact equipment enclosure, it is envisioned that the configurations and various techniques disclosed herein may be applied to other panels of an enclosure, such as a rear panel, a top panel, and so on. It is further understood that the various configurations and techniques disclosed herein may be applied to one panel, or to a combination of more than one panel, e.g., a front and rear panel.

[0043] Referring next to FIG. **1C**, a section view is provided illustrating a front portion **150** of the example of the electronic device **100** (FIG. **1A**) accordance with various aspects described herein. In particular, the section view is taken along line A-A' of FIG. **1A**. According to the illustrative example, a top, bottom and side panels of an equipment enclosure **140** are attached to a peripheral region of the front panel **102** as illustrated in part by the side panel **154** and the upper and lower panels **156a**, **156b** of the equipment enclosure **140** (FIG. **1B**). An interior region of the equipment enclosure **140** includes a main PCB **162**, shown in partial cross section. The main PCB **162** is planar and arranged along a mid-portion of the equipment enclosure **140**, e.g., extending from approximately an interior surface **153** of the front panel **102** towards a rear panel, not shown.

[0044] The illustrative embodiment also includes at least one upper chassis module **158** extending from a top surface of the main PCB **162** to approximately an interior surface of the upper panel **156a**. Likewise, the illustrative embodiment includes a lower chassis module **160** extending from a bottom surface of the main PCB **162** to approximately an interior surface of the bottom panel **156a**. In at least some embodiments, the upper and/or lower chassis modules **158**, **160** are attached to respective abutting surfaces of the main PCB **162**, e.g., being mounted thereon according to a mechanical and/or electrical fastening mechanism. The fastening mechanism may include one or more of a mechanical fastener, such as a rivet, a screw, an interference fit, a pin-socket arrangement, a solder joint, a weld, a chemical fastener, e.g., an adhesive, such as a glue and/or an epoxy, and the like.

[0045] According to the illustrative embodiment, the upper and lower chassis modules **158**, **160** extend to and/or at least partially through an adjacent region of the front panel **102**. The example chassis modules **158**, **160** may include, without limitation an electrical connector, a communication port, a heatsink and/or a vent. Accordingly, an upper reserved area **104a** accommodates a

corresponding portion of the upper chassis module **158**, while a lower reserved area **104b** accommodates a corresponding portion of the lower chassis module **160**.

[0046] The main PCB **162** defines a leading edge **163** arranged adjacent to a central region of the interior surface **153** of the front panel **102**. The leading edge **163** may be substantially close, e.g., in near intimate contact the interior surface **153** of the front panel **102**, but allowing for a relatively small gap **g**. It is understood that the gap **g** may be somewhat exaggerated in size for illustrative purposes. Nevertheless, the gap **g** may be sufficient to accommodate at least a portion of a status indicator assembly **164**, also shown in cross section. In at least some embodiments, the status indicator assembly **164** includes a status indicator abutting and/or extending at least partially into the interior surface **153** of the front panel **102**. The status indicator assembly **164** may be in communication with a status indication controller **166**, which may be located within the equipment enclosure **140**, e.g., on the main PCB **162**. The status indication controller **166** may provide one or more control signals adapted to selectively actuate one or more of the visual status indicators **112** (FIG. **1A**). Alternatively, or in addition, the front panel **102** may include a transmission device **168**, such as an acoustic and/or optical waveguide, configured to direct a status indication expressed by a status indicator **170** of the status indicator assembly **164** towards an exterior surface of the front panel **102**, such that expression of the status indication may be perceptible by a human observer.

[0047] FIG. **2A** is a perspective view illustrating a panel-facing surface of an example of a flexible status indicator assembly **200** in accordance with various aspects described herein. The flexible status indicator assembly **200** includes a substrate **202** having an elongated, substantially narrow and thin configuration. The substrate **202** may extend for a length that is many times its width, e.g., having a length that is up to 10, 20, 30, 50 or even 100 or more times its width. Similarly, the substrate may have a thickness that is substantially less than its width, e.g., having a thickness that is up to 10, 20, 30, 50 or even up to 100 or more times thinner than its width. In at least some embodiments, the substrate **202** is flexible. For example, the substrate **202** may be pliable, conformable, bendable and/or foldable. In at least some embodiments the substrate **202** is bendable to a minimum dimension, e.g., a minimum bend radius $r_{sub.min}$. In at least some embodiments, the minimum dimension may be determined according to a material and/or construction of the substrate **202** and/or according to a configuration of the substrate, e.g., having one or more different layers and/or according to circuitry supported by the substrate, e.g., conductive etches that may reside on one or more layers. Bends to radii greater than $r_{sub.min}$ may be accomplished without concern, whereas bends less than $r_{sub.min}$ may result in operational impacts and/or device damage, e.g., breakage. Alternatively, or in addition, the substrate **202** may be rigid, or substantially rigid, e.g., having a contour adapted to extend around an edge portion of a main PCB **162** (FIG. **1C**). In at least some embodiments, the substrate **202** may be configured with different segments having different mechanical properties along its length, such as a first segment that is substantially rigid and a second segment that is flexible, pliable, bendable, foldable and/or otherwise conformable.

[0048] In more detail, the example flexible status indicator assembly **200** includes an array of status indicator elements **204**. The status indicator elements **204** may include, without limitation, any combination of the various examples provided herein. The flexible status indicator assembly **200** further includes a terminal end, e.g., having one or more electrical contacts. In at least some embodiments, the contacts are provided in the form of a connector **203**. The flexible status indicator assembly **200** further includes an electrical circuit **208**. In at least some embodiments, the electrical circuit **208** includes one or more electrical circuits, electrical leads and/or conductive traces extending between at least one of the status indicator elements and the connector **203**.

[0049] In some embodiments, the electrical circuit **208** includes at least one conductive trace for each status indicator element **204**. However, as the number of status indicator elements **204** mounted on a single substrate **202** may be significant, e.g., 10, 20, 30, 50 or 100 or more, the electrical circuit **208** may be arranged to provide independent control of each status indicator

element **204** in an efficient manner by using fewer conductive traces than number of status indicator elements **204**. A ratio of status indicator elements **204** to conductive traces of such efficient designs may be substantial, e.g., 2:1, 4:1, 10:1 or more.

[0050] In at least some embodiments, the status indicator elements **204** of a common substrate **202** may be arranged according to multiple status indicator arrays and/or groups **206**. In at least some embodiments, the status indicator groups may be arranged along the length of the flexible status indicator assembly **200** according to a spatial multiplexing scheme. For example, a location of a status indicator group **206** may correspond with a proximal device, e.g., an adjacent electrical connector and/or telecommunications port.

[0051] FIG. 2B is a section view of the example of a flexible status indicator assembly **220** illustrated in FIG. 2A in accordance with various aspects described herein. The sectional view is taken along line B-B' (FIG. 2A) and includes a cross section of a portion of the flexible status indicator assembly **220**. The flexible status indicator assembly **220** includes a flexible substrate **221** that may include one or more distinguishable layers. According to the illustrative example, the flexible substrate **221** includes four layers **222a**, **222b**, **222c**, **222d**, generally **222**. In at least some embodiments, at least some of the layers include conductive traces of the electrical circuit **208** (FIG. 2A). An example status indicator element **224** is also shown in cross section, mounted to an outer surface of the flexible substrate **221**. It is understood that the status indicator element may include one or more electrical interconnections to conductive traces of one or more of the layers **222**.

[0052] The example flexible status indicator assembly **220** is positioned in alignment with a front facing edge of a main PCB **228**. In at least some embodiments, a substrate width w (FIG. 2A) of the substrate **202** may be equal to, slightly larger and/or slightly narrower than a thickness T of the main PCB **228**. For example, the width w may be less than about $5T$, or less than about $3T$, or less than about $2T$, or approximately T . To the extent a portion of the front panel is excluded from designation as a reserved space, e.g., along a front edge of the main PCB **228**, at least a portion of that excluded region may be utilized by the flexible status indicator assembly **220**.

[0053] In at least some embodiments, the flexible status indicator assembly **220** includes a fastening mechanism **225** adapted to secure at least a portion of the flexible status indicator assembly **200** with respect to one or more of the equipment enclosures **140** (FIG. 1C), the front panel **102** and/or a forward edge **223** of the main PCB **228**. The fastening mechanism **225** may include, without limitation, one or more of a mechanical fastener, a magnetic fastener, a chemical fastener and/or a weld. Examples of mechanical fastener include, without limitation, a pin, a rivet, a screw, a staple, a tack, an interference connector, and the like. Examples of chemical fasteners include, without limitation, an adhesive, a glue, an epoxy, and the like. In at least some embodiments, operation of the fastening mechanism **225** is reversible. Namely, the flexible status indicator assembly **220** may be attached, separated and reattached, e.g., to facilitate service and/or repair and replace activities.

[0054] FIG. 2C is an exploded plan view of a front corner portion of an electronic device **240** in accordance with various aspects described herein. The illustrated portion of the electronic device **240** includes a flexible status indicator assembly **242** and a portion of a main PCB **248**. In particular, the exploded view includes a top edge of the flexible status indicator assembly **242** in proximity to the portion of a main PCB **248**. According to the illustrative embodiment, the flexible status indicator assembly **242** includes an elongated structure having a panel segment **243a** at one end and a side segment **243b** at another end, with both segments **243a**, **243b** joined by a contoured segment **243c**. In this embodiment, the contoured segment includes a radius that conforms to a radial edge **249c** of the main PCB **248**. Although a radial edge **249c** is illustrated, it is understood that other shapes may be utilized, such as arcs, e.g., parabolic arcs, or more generally, any curvilinear shape, and/or linear shape, and/or combination of curvilinear and linear shapes.

[0055] In at least some embodiments, a shape of the corner of the main PCB **249** may be

determined according to a shape of the flexible status indicator assembly **242**, e.g., according to a minimum bend radius $r_{\text{sub.min}}$ of the flexible status indicator assembly **242**. Accordingly, the radial edge **249c** may have a radius equal to and/or slightly larger than the minimum bend radius. The illustrated portion of the main PCB **248** also includes a panel facing edge **249a** and a side facing edge **249b** joined by the radial edge **249c**. It is understood that a size, shape and/or contour of the corner of the main PCB **248** may impose restrictions upon placement of electrical, mechanical and/or electromechanical components thereon. For example, a radius of the radial edge **249c** may impose a restriction as to how much surface area of the front panel **102** (FIG. 1A) may be available for reservation. It is appreciated that any such restrictions are undesirable, such that shapes, radii and/or contours may be selected to optimize utilization of panel space.

[0056] FIG. 3 is a more detailed elevation view illustrating an example of a front panel **300** of another example electronic device in accordance with various aspects described herein. At least some electronic devices, such as telecom products, include multiple communication ports along with an arrangement of visual indicators, e.g., LEDs, associated with the ports. For example, at least one LED indicator may be associated with each port. The front panel **300** includes a first group of ports **302** that includes a quantity of eighty small-form-factor, 112 giga-bit-per-second (SFP112) ports **304**, each capable of dissipating about 4 Watts. The front panel **300** includes a first group of ports **302** that includes a quantity of eight quad-small-form-factor, double-density (QSFPDD) ports **309**, each capable of dissipating about 28 Watts.

[0057] The first group of ports **302** are arranged in four horizontal rows, with each row containing twenty ports. According to the example configuration, the first group of ports **302** can be considered as twenty, vertical, four-port stacks **306** distributed along a horizontal axis of the front panel **300**. Other areas of the front panel **300** in and around the first group of ports, includes openings or vents **320a** that may be used to allow cooling air to enter and/or exit the electronic device through the front panel **300**. The QSFPDD ports **309** are arranged in two groups of four. A lefthand group of ports **310a** is located between a left end of the first group of ports **302** and a left side of the front panel **300**. Similarly, a righthand group of ports **310b** is located between a right end of the first group of ports **302** and a right side of the front panel **300**. Other areas of the front panel **300** in and around the first group of ports **302**, include a first area openings or vents **320a** that may be used to allow cooling air to enter and/or exit the electronic device through the front panel **300**. Similarly, still other areas of the front panel **300** in and around the second groups of ports **310a**, **310b**, may include second areas with openings or vents **320b** that, once again, may be used to allow cooling air to enter and/or exit the electronic device through the front panel **300**.

[0058] Many times, a form factor of a product imposes constraints on a number of network ports that may be accommodated. In this instance, optical ports may be accommodated via the front panel **300**. A high number of ports, such as the example having eighty-eight ports, would necessarily occupy a substantial area of the front panel **300**. As these optical ports may include internal electronic devices, such as high-speed transceivers, they may consume a significant amount of power, consequently dissipating a substantial amount of heat in a relatively small, confined region. It may be appreciated that it may be important to control operational temperatures in such applications. It is understood that in at least some embodiments, the number and/or size of openings, e.g., vents, on the front panel **300** may be maximized to provide a greatest area possible area for heat dissipation and/or cooling air transfer. FIG. 3 also includes a drawing inset **311** that provides an enlarged view of a central region **313** of a portion of the front panel **300**. The central region **313** includes an arrangement of visual indicators in the form of a linear group of status indicators **312** including eighty LEDs **316**. Each LED **316** provides status and configuration information of a respective port of the example telecom device. The LEDs **316** may be arranged in groups **314**. Each group **314** includes four LEDs **316** associated with four adjacent SFP112 ports **304**, e.g., first and second LEDs **316** of the group **314** being associated with a top pair of the SFP112 ports **304** while third and fourth LEDs **316** of the group **314** are associated with a bottom

pair of the SFP112 ports **304**. A size of the group **314** may be determined according to a dimension of an associated port, e.g., a width of the SFP112 port **304**. Such group confinement restrictions may be used to spatially multiplex the LEDs **316**, to facilitate an association between the LEDs **316** and associated SFP112 ports **304** to a human observer.

[0059] In consideration of the respective areas of the front panel **300** being maximized for numbers of ports **304**, **309** and vents **320a**, **320b**, any remaining locations on the front panel **300** for accommodation of the status indicators **312** is limited. In such scenarios, it is not feasible to have traditional way of accommodating LEDs on faceplate. The central region **313** extends along a front edge of an interior main PCB (not shown). Accordingly, the central region **313** would be unavailable to accommodate any ports **304**, **309** and offering no apparent benefit for additional cooling vents, as the interior of the central region **313** is substantially blocked by an edge of the main PCB. Such a space may referred to as an exclusionary space, at least in that it is excluded from consideration of accommodating ports and/or cooling. Beneficially, however, the techniques disclosed herein provide a specially configured and/or operated status indicator assembly adapted for installation and/or operation on such exclusionary spaces.

[0060] FIG. **4A** is an elevation view illustrating a more detailed view of an example of a faceplate or front panel **400** of the example electronic device illustrated in FIG. **3**, equipped with space-saving visual indicators in accordance with various aspects described herein. The front panel **400** includes twenty quad port stacks **406**, each quad port stack including four SFP112 ports **404a**, **404b**, generally **404** arranged in vertical alignment. FIG. **4A** also includes a drawing inset **405** that illustrates an example quad port stack **406** in more detail. The quad port stack **406** includes an upper quad port pair **406a** and a lower port pair **406b**, with all four ports **404** arranged in vertical alignment. Also show are ventilation features of the front panel **400**. Namely, a first vent **420** provides cooling air access in regions of the front panel **400** that include an are above the upper quad port pair **406a** and below the lower quad port pair **406b**. In at least some embodiments, a second vent **421** provides cooling air access in regions of the front panel **400** located between stacked ports **404** of the upper quad port pair **406a** and/or between stacked ports **404** of the lower quad port pair **406b**.

[0061] It may be appreciated that areas of the front panel **400** including ventilation features, e.g., louvers, vents, screens and/or air filters, may not include status indicators, as such indicators would occupy areas of the front panel **400** dedicated to cooling. In such high-density and/or high-power applications, cooling areas are preferably maximized. It is observed that the illustrative front panel **400** includes a central region **413** located between the upper and lower quad port pairs **406a**, **406b**, generally **406**. The central region **413** may be narrow and is not allocated to either of the vents **420**, **421**. As such, the central region **413** presents an opportunity for hosting status indicators.

According to the example embodiment, the central region **413** is configured to accommodate an arrangement of status indicators associated with the quad port pairs **406**.

[0062] FIG. **4B** provides a perspective view illustrating an interior portion of a telecom device **450** utilizing the front panel **400** illustrated in FIG. **4A** with the front panel **400** removed in accordance with various aspects described herein. The telecom device **450** includes a central PCB **452**. The central PCB **452** is horizontally located in a middle region of the telecom device **450** and supports a first array of twenty upper transceiver pairs **454a**, stacked in pairs and mounted to an upper surface along a front edge of the central PCB **452**. The transceiver pairs **454a**, **454b**, generally **454** are associated with the SFP112 ports **304** (FIG. **3**). The central PCB **452** also supports a second array of twenty lower transceiver pairs **454b**, stacked in pairs and mounted to a lower surface along the front edge of the central PCB **452**. The upper and lower transceiver pairs **454a**, **454b** are vertically aligned to form twenty quad transceiver stacks, with the upper and lower mounting arrangement sometimes referred to as a “belly-to-belly” configuration. High-power transceivers associated with the QSFPDD ports **309** (FIG. **3**) are similarly stacked in vertical, belly-to-belly configurations, with a first high-power transceiver group **456a** located along a left end of the front edge of the central

PCB **452** and a second high-power transceiver group **456b** located along a right end of the front edge of the central PCB **452**.

[0063] FIG. **4B** also includes a drawing inset **460** that illustrates a first group of outer heatsinks **462a** located along an upper surface of the upper transceiver pair **406a** and a second group of outer heatsinks **462b** located along a lower surface of the lower transceiver pair **406b**. Similarly, a first group of inner heatsinks **464a** located in between transceivers of the upper transceiver pair **406a** and a second group of inner heatsinks **464b** located in between transceivers of the lower transceiver pair **406b**. It may be appreciated that there may be insufficient space to accommodate status indicators and/or light-pipes used in combination with internal LED indicators either above the upper transceiver pair **406a**, below the lower transceiver pair **406b**, because these areas are occupied by the outer heatsinks **462a**, **462b**. Likewise, there insufficient space to accommodate status indicators and/or light-pipes used in combination with internal LED indicators in between either of above the upper transceiver pair **406a** or the lower transceiver pair **406b**, because these areas are occupied by the inner heatsinks **464a**, **464b**. What space remains available in proximity to the quad port pairs **406** is the narrow space **463** located in between the upper transceiver pair **406a** and the lower transceiver pair **406b**. This space aligns substantially with the front edge of the central PCB **452**. Accordingly, space availability within an interior region of the telecom device **450** is limited.

[0064] FIG. **5** is a more detailed, exploded perspective view illustrating an interior portion of an example telecom device **500**, according to the devices **100**, **140**, **150**, **200**, **20**, **240**, **300**, **400**, **450** illustrated in FIGS. **1A-1C**, FIGS. **2A-2C**, FIG. **3** and FIGS. **4A-4B** in accordance with various aspects described herein. The telecom device **500** includes a main PCB **502** having an arrangement of network transceivers **504** attached to the main PCB **502** and aligned along a front edge of the main PCB **502**. The telecom device **500** includes a first status indicator assembly **506a** providing indications of status and/or configuration information of an associated first subset of the arrangement of transceivers **504**. Likewise, the telecom device **500** includes a second status indicator assembly **506b** providing indications of status and/or configuration information of an associated second subset of the arrangement of network transceivers **504**. According to the example illustration, the first status indicator assembly **506a** is associated with a lefthand portion of the arrangement of network transceivers **504**, while the second status indicator assembly **506b** is associated with a righthand portion of the arrangement of network transceivers **504**. A first drawing inset **510** illustrates in more detail, a segment of the first status indicator assembly **506a** including a flexible printed circuit **512** and a group of LED indicators **514**. A second drawing inset **530** illustrates in more detail, a proximal portion **531** of the first sticker LED PCBA **506a** a first terminal end **516a** of the first status indicator assembly **506a**. A second terminal end **516b** of the second status indicator assembly **506b** may be configured in a similar manner. Each of the first and second terminal ends **516a**, **516b** is positionable and interconnectable to a corresponding electrical connector **518a**, **518b**. The electrical connectors **518a**, **518b**, generally **518**, in turn, are in communication with a status indication controller **520**.

[0065] In at least some embodiments, the status indicator assembly **506a**, **506b**, generally **506**, includes an adhesive backed PCB **512**, e.g., a sticker LED PCBA (Printed Circuit Board Assembly) **512**. The sticker PCBA **512** may adhere to and/or be otherwise stuck across a front edge of the main PCB **502** thickness using a suitable adhesive. In some embodiments, a single sticker PCBA **512** may accommodate all eighty LEDs **514**. Alternatively, or in addition, multiple sticker PCBAs **512** may be utilized, with each sticker PCBA accommodating a subset of the total number of LEDs **514**. For example, a first sticker LED PCBA **506a** may be configured with forty LEDs **514**, while a second sticker LED PCBA **506b** may be configured with another forty LEDs **514** to collectively cover all eighty SFP112 port LEDs **514**. According to the illustrative example, the first sticker PCBA **506a** is associated with forty ports of a left half of the arrangement of transceivers **504**, while the second sticker PCBA **506b** is associated with forty ports of a right half of the

arrangement of transceivers **504**. In at least some embodiments, the first and second sticker PCBAs **506a**, **506b** have a common configuration. Accordingly, duplicates of the same PCBA **506** may be used for each of the left half and the right half of the arrangement of transceivers **504**, e.g., by assembling the status indication system according to a mirrored configuration. Since this sticker PCBA **506a** is mounted across a thickness of the PCB **502**, it does not obstruct air flow. Each of the example sticker PCBAs **506** terminate to a controller module, e.g., to the status indication controller **520**, e.g., located on the main PCB **502** via respective connectors **516a**, **516b**, generally **516**. For flexible PCB applications, it is envisioned that the connectors may include flex-type connectors to simplify design, construction and/or costs.

[0066] FIG. **6A** provides an elevation view illustrating a front panel **602** of another example electronic device **600** equipped with space-saving visual indicators in accordance with various aspects described herein. The electronic device **600** includes an equipment enclosure or housing **654** (FIG. **6B**) that includes an exposed panel **602** that may be positioned for observance by a human observer. The exposed panel **602** includes an arrangement of status indicators **608** including a number of individual status indicators **612** configured to provide configuration and/or status information to a human observer. The exposed panel **602** further includes a group of reserved areas **604a**, **604b**, **604c**, generally **604**, which may be designated as reserved because they are unavailable to accommodate the arrangement of status indicators **608**. One or more of the reserved areas **604** may be unavailable because it is occupied by another device, such as an electrical connector and/or an electromechanical port that would preclude positioning of a status indicator **612** there. Alternatively, or in addition, one or more of the reserved areas **604** may be unavailable because it is occupied by a vent, or ventilation panel as may be used for intake and/or exhaust cooling air flow. In at least some embodiments, one or more of the reserved areas **604** may be unavailable because it is occupied by a design element, such as a label and/or branding mark. According to the foregoing examples, unavailability is due to the existence of a structure that may be observable from an observer of the front panel **602**.

[0067] It is conceivable that in at least some embodiments, unavailability may be due to a condition occurring on an interior portion of the device **600**, e.g., behind the front panel **602**. Such conditions may include, without limitation, a location of an electrical, mechanical and/or electromechanical assembly, device and/or component within an interior region of the device **600** and proximate to an interior surface of the front panel **602**. The condition may arise from a mechanical interference, e.g., insufficient room to accommodate the status indicator **612**, and/or some other reason, such as a possibility of thermal interference and/or electrical interference, e.g., electromagnetic interference. It is understood that in such instances, the internal condition may not be visible by an observer of the front panel **602** but may nevertheless preclude positioning of a status indicator **612** there.

[0068] In at least some embodiments, the front panel **602** may include one or more exclusion regions that may be unavailable for reservation according to the example reserved areas **604**. It is understood that in at least some embodiments, a panel surface area may include one or more reserved areas and one or more unreserved areas, in which unreserved areas are so designated by their unavailability for serving as reserved areas. By way of example, such exclusion regions may include at least a portion of a peripheral region of the front panel **602** as may be occupied by structural supports, e.g., for attachment to top, bottom and/or side panels, and/or internal structures such as mounting frames and/or printed circuit boards. The example front panel **602** includes a lower exclusion region **610** extending between a bottom perimeter of the front panel **602** and a bottom edge of the lower reserved areas **604**. The lower exclusion region **610** may be unavailable for reservation due to the location of an adjacent edge of a PCB, e.g., a main PCB or motherboard. It is understood that in at least some embodiments, one or more of any identified exclusion regions, e.g., the lower exclusion region **610**, may be available to accommodate one or more of the status indicators **612**, despite being unavailable to accommodate a reserved area **604**. Indeed, at least one strategy that supports maximizing reserved area(s) **604**, while also providing status

indicators **612**, takes advantage of locating at least a portion of the status indicators within the exclusion regions.

[0069] Referring next to FIG. **6B**, a section view is provided illustrating a front portion **650** of the example electronic device **600** illustrated in FIG. **6A** in accordance with various aspects described herein. In particular, the section view is taken along line C-C' of FIG. **6A**. An equipment chassis **654** is attached to a peripheral region of the front panel **602** as illustrated in part by upper and lower surfaces **656a**, **656b** of the equipment chassis **654**. The electronic device **600** includes a main PCB **662**, shown in partial cross section. The main PCB **662** is planar and arranged along a lower portion of the equipment chassis **654**, e.g., extending from approximately an interior surface **653** of the front panel **602** towards a rear panel, not shown.

[0070] The illustrative embodiment includes a chassis module **658** extending from a top surface of the main PCB **662** to approximately an interior surface of the upper panel **656a**. In at least some embodiments, the chassis module **658** is attached to a respective abutting surface of the main PCB **662**, e.g., being mounted thereon according to a mechanical and/or electrical fastening mechanism. The fastening mechanism may include one or more of a mechanical fastener, such as a rivet, a screw, an interference fit, a pin-socket arrangement, a solder joint, a weld, a chemical fastener, e.g., an adhesive, such as a glue and/or an epoxy, and the like.

[0071] According to the illustrative embodiment, the chassis module **658** extends to and/or at least partially through an adjacent region of the front panel **602**. Example chassis modules **658** include, without limitation, an electrical connector, a communication port, a heatsink and/or a vent. Accordingly, a reserved area **604a** accommodates a corresponding portion of the chassis module **658**.

[0072] The main PCB **662** defines a leading edge **663** arranged adjacent to a lower region of the interior surface **653** of the front panel **602**. The leading edge **663** may be substantially close, e.g., in near intimate contact the interior surface **653** of the front panel **602** but allowing for a relatively small gap. It is understood that the gap may be somewhat exaggerated in size for illustrative purposes. Nevertheless, the gap may be sufficient to accommodate at least a portion of a status indicator assembly **664**, also shown in cross section. In at least some embodiments, the status indicator assembly **664** includes a status indicator abutting and/or extending at least partially into the interior surface **653** of the front panel **602**. The status indicator assembly **664** may be in communication with a status indicator controller (not shown), which may be located within the equipment chassis **654**, e.g., on the main PCB **662**. The status indicator controller may provide one or more control signals adapted to selectively actuate one or more of the status indicators **670**. Alternatively, or in addition, the front panel **602** may include a transmission device **668**, such as an acoustic and/or optical waveguide, configured to direct a status indication expressed by a status indicator **670** of the status indicator assembly **664** towards an exterior surface of the front panel **602**, such that expression of the status indication may be perceptible by a human observer.

[0073] FIG. **7A** is an elevation view illustrating a front panel of yet another example electronic device **700** equipped with space-saving visual indicators in accordance with various aspects described herein. The electronic device **700** includes a housing **754** (FIG. **7B**) that includes an exposed panel **702** that may be positioned for observance by a human observer. The exposed panel **702** includes an arrangement of status indicators **708** including a number of individual status indicators **712** configured to provide configuration and/or status information to a human observer. The exposed panel **702** further includes a reserved area **704**, which may be designated as reserved because it is unavailable to accommodate the arrangement of status indicators **708**. One or more of the reserved areas **704** may be unavailable because it is occupied by a vent, or ventilation panel **753** (FIG. **7B**) as may be used for intake and/or exhaust cooling air flow. According to the foregoing examples, unavailability may be due to the existence of a structure that may be observable from an observer of the front panel **702**.

[0074] In at least some embodiments, the front panel **702** may include one or more exclusion

regions that may be unavailable for reservation according to the example reserved areas **704**. By way of example, such exclusion regions may include at least a portion of a peripheral region of the front panel **702** as may be occupied by structural supports, e.g., for attachment to top, bottom and/or side panels, and/or internal structures such as mounting frames and/or printed circuit boards. The example front panel **702** includes a lower exclusion region **710** extending between a bottom perimeter of the front panel **702** and a bottom edge of the lower reserved area **704**. The lower exclusion region **710** may be unavailable for reservation due to the location of an adjacent edge of a PCB, e.g., a main PCB or motherboard. It is understood that in at least some embodiments, one or more of any identified exclusion regions, e.g., the lower exclusion region **710**, may be available to accommodate one or more of the status indicators **712**, despite being unavailable to accommodate the reserved area **704**. Indeed, at least one strategy that supports maximizing reserved area **704**, while also providing status indicators **712**, takes advantage of locating at least a portion of the status indicators within the exclusion regions.

[0075] Referring next to FIG. 7B, a section view is provided illustrating a front portion **750** of the example electronic device illustrated in FIG. 7A in accordance with various aspects described herein. In particular, the section view is taken along line D-D' of FIG. 7A. An equipment chassis **754** is attached to a peripheral region of the front panel **702** as illustrated in part by upper and lower surfaces **756a**, **756b** of the equipment chassis **754**. The electronic device **700** includes a main PCB **762**, shown in partial cross section. The main PCB **762** is planar and arranged along a lower portion of the equipment chassis **754**, e.g., extending from approximately an interior surface **753** of the front panel **702** towards a rear panel, not shown.

[0076] The illustrative embodiment includes a chassis module **768** attached to a top surface of the main PCB **762**. In at least some embodiments, the chassis module **758** is attached to a respective abutting surface of the main PCB **762**, e.g., being mounted thereon according to a mechanical and/or electrical fastening mechanism. The fastening mechanism may include one or more of a mechanical fastener, such as a rivet, a screw, an interference fit, a pin-socket arrangement, a solder joint, a weld, a chemical fastener, e.g., an adhesive, such as a glue and/or an epoxy, and the like.

[0077] According to the illustrative embodiment, the reserved area **704** includes a cooling vent **753**. The device **700** also includes a cooling fan **772** adapted to move cooling air through the cooling vent **753**. The cooling air may be adapted to provide thermal management for the device **700**, e.g., for the chassis module **758**.

[0078] The main PCB **762** defines a leading edge **763** arranged adjacent to a lower region of the interior surface **753** of the front panel **702**. The leading edge **763** may be substantially close, e.g., in near intimate contact the interior surface **753** of the front panel **702** but allowing for a relatively small gap. It is understood that the gap may be somewhat exaggerated in size for illustrative purposes. Nevertheless, the gap may be sufficient to accommodate at least a portion of a status indicator assembly **764**, also shown in cross section. In at least some embodiments, the status indicator assembly **764** includes a status indicator abutting and/or extending at least partially into the interior surface **753** of the front panel **702**. The status indicator assembly **764** may be in communication with a status indicator controller (not shown), which may be located within the equipment chassis **754**, e.g., on the main PCB **762**. The status indicator controller may provide one or more control signals adapted to selectively actuate one or more of the status indicators **770**. Alternatively, or in addition, the front panel **602** may include a transmission device **768**, such as an acoustic and/or optical waveguide, configured to direct a status indication expressed by a status indicator **770** of the status indicator assembly **764** towards an exterior surface of the front panel **702**, such that expression of the status indication may be perceptible by a human observer.

[0079] According to a space-saving configuration, an area allocated to any of the example status indicator assemblies is very limited. It is understood that in at least some embodiments, the status indicator assemblies include at least one electrical trace and/or circuit formed on and/or within a substrate material. Substrate materials may include any suitable substrate to support printed

circuits, such as rigid materials, bendable materials, formable materials, flexible materials and/or foldable materials. In at least some embodiments, the substrate material may include substantially insulative properties so as not to interfere with operation of any electrical signals carried by the electrical circuits and/or traces. Alternatively, or in addition, in at least some embodiments, dimensions, e.g., thickness and/or material properties, e.g., dielectric constant, electrical conductivity, resistivity, permittivity, magnetic permeability, thermal conductance, and the like may be selected according to a design process for the status indicator assembly.

[0080] For example, space on an example sticker LED PCBA **506** (FIG. 5) to carry signals may be limited by the thickness of the main PCB **502** (FIG. 5), vertical spacing between a port and a surface of the main PCB **502**, and so on. In one illustrative example, the LEDs **514** are tri-color LEDs, e.g., having separately controllable red, green and blue LED components. Continuing with the illustrative example, forty tri-color type LEDs would require at least **120** signals, one for each color dimension of each LED of the forty LEDs. In all likelihood, a power plane and/or a ground plane or signal return may also be required. Considering that each LED color dimension typically consumes about 10 Milliampere (mA) when activated, a supporting status indicator circuit of one of the example sticker LED PCBAs **506** would be required to carry and/or otherwise supply a maximum current of about 1.2 Amps (A). Thus, a conventional approach would require 120 signals, e.g., 120 signal traces, and a power plane configured to supply up to 1.2 A of current.

[0081] It should be appreciated that a form factor of a flexible PCBA, such as the example sticker LED PCBA **506**, may not be well suited for such high densities of control signals and/or operating currents. The illustrative embodiments disclose devices, processes and control software configured to reduce the number of signals and/or circuits and/or conductive PCB traces without compromising functionality. Namely, the same number of status indicators, e.g., LEDs, may be controlled to provide indications of status and/or configurations of each of the ports.

[0082] FIG. 8 provides a schematic diagram of an example status indicator system **800** incorporating the example status indicator assembly of FIGS. 2A-2C in accordance with various aspects described herein. The status indicator system **800** includes a respective status indicator, e.g., an LED **804**, for each of a number of ports, e.g., forty ports, referred to as Port_01, Port_02, . . . Port_40. According to the illustrative example, each port includes a tri-colored LED **804**. At least some of the example tri-colored LEDs **804** include a red LED **806a**, a green LED **806b** and a blue LED **808c**. The LEDs **806a**, **806b**, **806c**, generally **806**, may be actuated independently to form a particular color of a range of colors. Actuation may include forward biasing a selected combination of the LEDs **806** with an operational voltage, e.g., VCC, having sufficient value to turn the selected combination of LEDs **806** on thereby providing illumination according to a specific color mix. By way of example, independent actuation of each of the independent red, green, blue LEDs **806** provides the corresponding red, green or blue color. However, actuation of a combination, such as the red and green LEDs **806a**, **806b** may produce a color mix yielding the color yellow. Similarly, actuation of all three colors, red, green and blue may produce a color mix yielding the color white. Other color combinations are possible.

[0083] The example status indicator system **800** further includes a status indication controller **810** that may be configured to provide selective actuation of the LEDs of each port according to a predetermined status and/or configuration of that port. For example, and without limitation, the color white may indicate the corresponding port is configured for operation in an idle state, the color green may indicate transmit mode operation, yellow may indicate receive mode operation, and red may indicate an error status or failure.

[0084] The status indication controller **810** may be in communication with electrical conductors **811** of a cable harness and/or traces of a PCB, such as the example flexible PCBs disclosed herein. In at least some embodiments, one or more interconnections may be provided between the status indication controller **810** and the electrical conductors **811**. The example status indicator system **800** includes a single, multi-contact connector **812** connecting inputs and/or outputs of the status

indication controller **810** to respective ones of the electrical conductors **811**. At least one operational voltage, e.g., VCC, may be provided by a power supply **815**. It is understood that application of a positive voltage VCC to an anode terminal of any one of the LEDs **806**, will turn the respective LED on when a circuit is completed through the LED **806**.

[0085] The LEDs **804** are arranged according to multiple groups of ports. According to the illustrative example, a first group of ports **808a** includes four ports, namely, Port_01, Port_02, Port_03, Port_04 **802a**, **802b**, **802d**, **802c**, generally **802**. The remainder of the forty ports may be arranged in groups in a like manner, resulting in a first group **808a**, second group, and so forth, through a tenth group **808b**. Each tri-colored LED **804** of the group may be in communication with a respective one of four actuation circuits, e.g., four VCC circuits **816a**, **816b**, **816c**, **816d**, generally **816**. Each group **808** may be configured in a like manner, such that an application of any one of the VCC circuits **816**, e.g., a first VCC circuit **816a**, applies a first voltage VCC to the first tri-colored LED **804** of each of the ten groups of ports **808a** through **808b**, generally **808**. It is understood that in at least some embodiments only one of the VCC circuits **816** may be active at any given time.

[0086] In at least some embodiments, the controller provides additional control signals to each of the groups of ports **808**. For example, a first red control signal **814a** is in communication with a cathode of each of the red LEDs **806a** of each of the four tri-color LEDs **804** of the first group of ports **808a**. Similarly, a first green control signal **814b** is in communication with a cathode terminal of each of the green LEDs **806b** of the four tri-color LEDs **804** of the first group of ports **808a**, and a first blue control signal **814c** is in communication with a cathode terminal of each of the blue LEDs **806c** of the four tri-color LEDs **804** of the first group of ports **808a**. Accordingly, actuation of one of the VCC circuits **816** selects one of the tri-color LEDs **804** associated with one of the ports of each of the multiple groups, e.g., ten tri-color LEDs **804** in a simultaneous manner. A particular color, or lack thereof may be controlled for the selected tri-color LED **804** according to control signals applied by the status indication controller **810** via the red, green and blue control signals **814a**, **814b**, **814c**. In at least some embodiments, each of the different groups receives a respective combination of red, green and blue control signals for that group.

[0087] According to the illustrative embodiment, the status indicator system **800** divides a set of port LEDs, e.g., ports 1-40, into groups of four ports, with each LED of the four port LEDs sharing a common control signal lead to the status indication controller **810**. The power signals, e.g., VCC_1 through VCC_4 may be driven by field effect transistors (FETs) **820**. For example, the status indication controller **810** may provide a switching signal to one or more of the FETs **820**, to turn the FETs **820** on selectively. When any of the FETs **820** are switched on, the power supply voltage, e.g., VCC, is applied to an interconnected power circuit **816**. It is envisioned that in at least some embodiments, the FETs **820** may be driven in a repetitive manner, or frame as described further below.

[0088] FIG. 9 illustrates a timing diagram **900** of example control signals of the example status indicator circuit **800** of FIG. 8 in accordance with various aspects described herein. A time axis T marks a reference time T.sub.0 and measures out subsequent time intervals according to frames. The frames, referred to as port frames **906**, are arranged sequentially along the timeline and numbered accordingly, with a first port frame **906a** occupying a time interval $T_{sub.0} \leq t \leq T_{sub.1}$, a second port frame **906b** occupies a time interval $T_{sub.1} \leq t \leq T_{sub.2}$, a third port frame **906c** occupies a time interval $T_{sub.2} \leq t \leq T_{sub.3}$ and a fourth port frame **906d** occupies a time interval $T_{sub.3} \leq t \leq T_{sub.4}$. According to a time division multiplexing scheme, each port frame is associated with a respective one of the power circuits **816** (FIG. 8).

[0089] In at least some embodiments, only one power circuit **816** is active during each port frame **906a**, **906b**, **906c**, **906d**, generally **906**. The power circuits **816** may sequence according to a predetermined order. For example, during the first port frame **906a** only the first power circuit **816a** is active, referred to as VCC_1, while all of the other power circuits **816b**, **816c**, **816d**, referred to

as VCC_2, VCC_3 and VCC_4 are inactive, e.g., disabled. Once again, whether a particular power circuit **816** is active may be determined by control signals from the status indication controller **810** (FIG. **8**) activating respective FETs **815**. When VCC_1 power circuit **816a** is enabled, the first port LED **804** (FIG. **8**) of each group **808a** through **808b** is controlled via their respective LED control signals **814**. All the other ports LEDs remain inactive, e.g., turned off as their supply, e.g., VCC, is disconnected or otherwise referred to as “off.” Accordingly, when VCC_1 power circuit **816a** is enabled, the control signals R1, G1 and B1 are applied as described above in relation to FIG. **8**, to control the individual LEDs **806** of the tricolor LED **804** of the first port of each group, e.g., Port_1 **802a** of Group_1 **808a**. Likewise, during the first port frame **906a**, a second group of control signals R2, G2 and B2 controls a tri-color LED of a first port of a second group, e.g., port5 (not shown), and so on for the other groups **808**. The tri-color LEDs **804** for all other ports will be off during the first port frame **906a**.

[0090] In next frame, e.g., a second port frame **906b**, the second VCC_2 power circuit **816b** is enabled. The first group of control signals R1, G1 and B1 are provided to control a tricolor LED of a second port of each group, e.g., Port_2 **802b** of Group_1 **808a**, the same second group of control signals R2, G2 and B2 controls are provided to control a tricolor LED of a second port of a second group, e.g., the tri-color LED of Port_6 LED and so on for each of the remaining groups. The status indication controller **810** may be adapted to apply respective control states for the groups of control signals R1, G1, B1, R2, G2, B2, etc., such that the tri-color LEDs of each port may be independently controlled.

[0091] The first sequence of port frames **906** may be grouped together according to a first port group frame **908a**, in which port frames **906** are not repeated. A second sequence of port frames **906** may be grouped together according to a second port group frame **908b**, in which, once again, port frames **906** are not repeated. In at least some embodiments, the port frames **906** of each of the sequential port group frames actuates the same ports according to the same sequence, e.g., representing a subsequent update of activation of a status indicator of a particular port. In this manner, the process may repeat indefinitely. It is understood that a particular indication of each status indicator, e.g., a particular color, may be controlled according to a condition of a corresponding port. Thus, if the port condition does not change, the same color may be expressed for that port during subsequent port group frames **908**. However, if the port condition changes, an updated color may be expressed for that port according to the changed condition.

[0092] A duration of the port frame **906** may be selected according to a predetermined time interval during which the corresponding port would be activated, e.g., lighting an LED status indicator. A number of port frames in a port group frame **908** may be determined according to a circuit sharing scheme in which a total number of status indicators is divided into multiple groups, each having fewer indicators than the total. For example, a group of forty tri-color indicators may be subdivided into ten groups of four. Accordingly, each time a port frame is activated, a corresponding port of each group is activated simultaneously. Consider a first port of each of the ten groups of ports being activated during a first port frame **906a**, while the second, third and fourth ports of each group remain inactive. Likewise, a second port of each of the ten groups of ports may be activated during a second port frame **906b**, while the first, third and fourth ports of each group remain inactive. The process may continue in a similar manner until all ports of each group of ports have been activated, upon which time the process may repeat.

[0093] In this manner, the status indicator circuit **800** provides a signal control strategy for connecting a relatively large number of ports, e.g., forty optical ports, each port configured with a multi-input device, such as single tri-color LED, using substantially fewer circuits and/or conductive paths than total number of ports. Additionally, a power savings may be realized as each port LED is controlled for a fraction of time, e.g., every 1 msec. If a particular LED is to be turned on, it is on for 250 μ Sec and off for 750 μ sec. Consequently, the LED is operated according to an on-off rate, or a “blink” rate of 1,000 Hz. Other switching rates and/or duty cycles are possible,

with an understanding that the switching rates and/or duty cycles should be sufficiently fast and/or of sufficient duration such that any on-off flickering remains imperceptible to the human eye. It is understood that the human eye may be unable to distinguish switching or blink rates of 25 Hz or greater. Accordingly, LED switching rates at or above about 25 Hz should appear to the human eye as though the LEDs remain on.

[0094] It is understood that numbers of ports, sizes of groups, duration of group frames, and so on may be determined according to design criteria. Example design criteria may include, without limitation, physical circuit constraints, e.g., maximum number of signal leads, power leads and/or power planes, numbers of PCB layers, equipment panel area sizes, equipment panel reserved area constraints, equipment area exclusion area constraints, power requirements, maximum flicker rates, and so on.

[0095] Table 1 provides a comparison with respect to input/output count required on a status indicator PCB assembly and/or connector. Table 1 also provides an example comparison of LED power consumption for a conventional approach versus the example time division multiplexing scheme of the foregoing example.

TABLE-US-00001 TABLE 1 Comparison of Time Division Multiplexing to Traditional Approach.

Advantage	Conventional	Proposed	Details	Scheme	Scheme	Digital Signals
$40 \times 3 = 120$	$3 \times 10 = 30$	—	Max Current*	$40 \times 3 \times 10$	$3 \times 10 \times 10$	—
Requirement	$\text{mA} = 1200$	$\text{mA} = 300$	Average Power	$3.3 \times 1200 = 3.96$	$3.3 \times 300 = 0.99$	W
3.3 V supply.	(V $\times$ I) Power saving = 75%	Power Contact	3 4	—	Total IO Count	$120 + 3 = 123$
+ 4 = 34	IO Count (Digital + Power) reduction by = 72.35%					

[0096] A maximum current requirement referred to in Table 1, may be determined according to a situation in which all the ports tri colors LEDs are activated or glowed, and each LED is sinking about 10 mA of current. The example signal reduction techniques may be implemented to drastically reduce a number of signals required to activate or otherwise light LEDs in different colors as per the desired color coding for the corresponding ports, e.g., the example optical ports.

[0097] By way of example, the status indicator control signal values and timing aspects may be represented in a matrix according to the timing diagram **900**. The LEDs may be activated and/or otherwise glowed, with columns of four separate power rails (VCC) **816** (FIG. **8**) and four rows of sharing LED control signals. The various different column and row signals may be controlled in a time-division-multiplexing (TDM) manner. This process may be repeated at a sufficiently high frequency to avoid human eye perceiving blinking of LEDs. Such a time division signal sharing allows for fewer control signals and/or power planes, which is beneficial to reduce size and/or power requirements of the status indicator PCB assembly, e.g., requiring fewer PCB layers, narrower conductive traces, and so on. . . .

[0098] According to the illustrative example, the overall power consumption of the LED circuit design may be reduced by a factor of four, requiring only about 25% of what would have otherwise been required for full, independent LED control without. Beneficially, resulting reductions in the size requirement, circuit complexity and power may be accomplished without sacrificing visual signal quality of the LEDs.

[0099] The techniques disclosed here provide space-saving status indicator circuits capable of operating large numbers of status indicator elements, such as LEDs, with shared power and/or LED control leads. The resulting status indicator assemblies may be used individually and/or in any combination for a particular electronic device. Solutions using multiple status indicator assemblies may use different size and/or shaped status indicator assemblies that may be configured to control the same and/or different numbers of status indicator elements. In at least some embodiments, similar and/or identical status indicator assemblies may be used to control different groups of status indicators, e.g., according to the left-right example illustrated in FIG. **5**. In at least some embodiments, the reduced size and/or power requirements allows for placement in areas that might otherwise be excluded and/or unavailable, such as along the edges of internal devices, such as the

example main PCBs. The example status indicator assemblies may be combined with simple attachment facilitating feature, mechanism and/or device, such as adhesives and/or magnets to permit the status indicator assemblies to be placed in permanent and/or removable manner, as in the illustrative example in which a flexible LED PCBA is combined with an adhesive back, facilitating attachment of the assembly across the thickness of a main board PCB. In this manner, the LED PCBA may be assembled close to an interior surface of the faceplate.

[0100] In at least some embodiments, an equipment panel or faceplate may incorporate openings, e.g., holes, through which a transmission device, such as a light pipe, may be routed, e.g., press fitted. Such arrangements places one end of a light pipe very close to a status indicator LED, while an opposing end of the light pipe is exposed along the equipment panel for external observation. Use of transmission devices, such as the example light pipes helps to prevent interference, e.g., color bleeding, that might otherwise be experienced from adjacent status indicators, e.g., LEDs. Beneficially, this approach minimizes requirements for additional space and/or does not occupy any additional space on a panel of an equipment enclosure. For example, the light pipes may include columns directing light perpendicularly from the LED to an exposed surface of the front panel. Such approaches facilitate maximum openings on the equipment panel for air inlet and/or exit to cool high power devices, such as the example optical network ports.

[0101] FIG. **10** depicts an illustrative embodiment of an equipment status indication process **1000** in accordance with various aspects described herein. The process **1000** facilitates operation of up to a substantial number of status indicators, e.g., LEDs, in a space-conserving manner. According to the examples provided herein, such a process conserves a significant amount of front panel space and/or internal space of a corresponding equipment housing that may be allocated to other purposes, such as cable interconnects, operator controls and/or cooling.

[0102] According to the example process, a group of status indicators distributed along a proximal end of an elongated flexible PCB assembly is identified at **1002**. Identification may include one or more of identifying a number of status indicators, a type and/or types of status indicators, a location of the status indicators, a grouping of the status indicators, and so on. Groupings may include division of a total number of panel status indicators into different groups serviced by respective elongated flexible PCB assemblies. As in the example of FIG. **5**, a total number of eighty LEDs **514** is divided into two groups supported by two flexible PCB assemblies **506a**, **506b**.

[0103] A status indicator control signal may be received at **1004**. Status indicator control signals may include one or more of an activation or actuation signal that may operate to selectively turn a status indicator on or off. A nonlimiting example includes the selective application of VCC to one of a number of subgroups of LEDs via the frames **906**. Alternatively, or in addition, the status indicator control signal may include a status indicator configuration control. A status indicator configuration control may include one or more of a color, an intensity, e.g., of the example tri-color LEDs and/or other status indicator features, such as a size, a shape, a tone, a volume, and so on.

[0104] In at least some embodiments, the status indicator control signal may be applied, at **1006**, to proximal end of the elongated flexible PCB assembly. According to the illustrative example, a status indicator controller **520** generates the status control signal and applies the signal to a proximal electrical interconnect, e.g., a connector **516a**, of the example elongated flexible PCB assembly **506a**. The applied control signals are directed, in turn, at **1008** along the elongated flexible PCB assembly to the group of status indicators. The status indicators may be actuated, at **1010**, according to status indication control signal. Beneficially, panel area may be conserved at **1012** by routing control signals via the elongated the flexible PCB assembly to panel space that might not otherwise be available for allocation to other front panel uses, such as cable interconnects, operator controls and/or cooling.

[0105] FIG. **11** depicts an illustrative embodiment of an equipment status indication control process **1100** in accordance with various aspects described herein. According to the example process **100**, one or more design constraints are identified at **1102**. Example design constraints may include,

without limitation, one or more of panel space of an equipment enclosure available to status indicators, maximum number of layers and/or available circuit routing width and/or area of a multi-layer PCB, such as the example flexible status indicator assemblies **200**, **506** (FIGS. 2A and 5) disclosed herein, maximum permissible thermal load, shape constraints, such as a maximum elongated PCB width, a PCB width to length ratio, a minimum bend radius, numbers and/or types of status indicators, spatial multiplexing for placement of status indicators, e.g., splitting a total number of status indicators into multiple status indicator sub-groups supported on multiple flexible PCBs of an equipment panel, such as the first and second (left and right) status indicator assemblies **506**, electrical requirements, e.g., maximum voltage and/or currents, and so on.

[0106] Continuing with the example process **1100**, a total number of status indicators may be divided at **1104** into multiple groups according to design constraint. For example, a large number of status indicators may be subdivided and/or otherwise grouped into multiple smaller groups that may be individually operated and/or controlled, e.g., in a sequential and/or simultaneous arrangement.

[0107] Status indicator control signals are generated, at **1106**, for number of status indicators. Status indicator control signals may include one or more of an activation or actuation signal that may operate to selectively turn a status indicator on or off. A nonlimiting example includes the selective application of VCC to one of a number of subgroups of LEDs via the frames **906**. Alternatively, or in addition, the status indicator control signal may include a status indicator configuration control. A status indicator configuration control may include one or more of a color, an intensity, e.g., of the example tri-color LEDs and/or other status indicator features, such as a size, a shape, a tone, a volume, and so on.

[0108] A lesser number of signal control leads is identified, at **1108**, to be shared by group members. For example, a signal sharing technique, such as a multiplexing arrangement may be used for control. Multiplexing may include one or more of temporal, frequency and/or spatial multiplexing to reduce a number of signal control leads and/or conductive traces for any given number of status indicators. Selection of the lesser number of signals may be determined, in at least some embodiments, according to one or more of the design constraints identified at **1102**. For example, a total number of status indicators may be subdivided into multiple groups, with each group supported by a respective flexible PCB assembly according to a spatial multiplexing scheme. Alternatively, or in addition, a total number of status indicators of any given flexible PCB may be subdivided further into sub-groups that may be controlled according to a multiplexing scheme, e.g., a time division multiplexing (TDM) scheme, in which each group may be activated or lit within a respective time slot to the exclusion of other groups, which may remain un-activated or dark during that time slot. Each sub-group may have an assigned activation period that may be revisited periodically according to the TDM scheme. In at least some embodiments, a group revisit time of the TDM scheme may be selected according to a tolerable on/off blink rate of the status indicators. Accordingly, the number and/or size of the sub-groups and/or an extent of a multiplexing scheme may be determined according to a maximum number of status indicators and/or a tolerable blink rate.

[0109] Other considerations of sub-group size may include physical constraints. For example, if an elongated flexible PCB assembly is to be routed around an edge of a structure, such as the example main PCB **248** (FIG. 2C), a minimum bend radius may be a function of the number of flexible PCB layers. Accordingly, in at least some embodiments, a minimum bend radius may be determined according to a space conserving constraint of front panel area. Namely, the lesser the bend radius, the greater the panel area is available to an access length of the main PCB **248**. The minimum radius may then be used to determine a maximum number of flexible PCB layers may be permissible for a given flexible PCB assembly architecture. The number of flexible PCB layers may then be used to determine a maximum number of supportable status indicators, e.g., according to one or more multiplexing techniques as may be applied. The various interrelations of parameters

may be solved in any suitable manner, including in optimal manners in which one or more of the parameters are identified for optimization, allowing the other parameters to be determined accordingly.

[0110] The status indicator control signals may be provided, at **1110**, to status indicators via shared signal control leads. For at least some embodiments in which the status indicators are subdivided into smaller groups, e.g., individual network ports, and/or groups of ports, such as the example quad port pairs **406** (FIGS. **4A**, **4B**), a member selection signal may be identified, at **1112**. The member selection signal may identify one or more of the subgroups and/or individual members of one or more sub groups, e.g., a first LED of each group of ports. Ultimately, the group member selection signal may be applied at **1114** to activate and/or otherwise control operation of selected group member according to status indicator control signals.

[0111] While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIGS. **10** and **11**, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

[0112] Turning now to FIG. **12**, there is illustrated a block diagram of a computing environment **1200** in accordance with various aspects described herein. In order to provide additional context for various embodiments of the embodiments described herein, FIG. **12** and the following discussion are intended to provide a brief, general description of a suitable computing environment **1200** in which the various embodiments of the subject disclosure can be implemented. In particular, computing environment **1200** can be used in the implementation of the status indication controller **166**, **520**, **810** (FIGS. **1**, **5** and **8**). Each of these devices can be implemented via computer-executable instructions that can run on one or more computers, and/or in combination with other program modules and/or as a combination of hardware and software. For example, computing environment **1200** can facilitate in whole or in part providing status indication by way of control signals at a proximal end of an elongated flexible circuit assembly that includes a distal portion attached to and extending along at least a portion of an elongated, panel-facing edge of a planar circuit board. The control signals may apply a multiplexing technique to reduce a number of signal leads to accommodate a relatively large number of status indicators. In at least some embodiments. The control signals may be directed around a bent portion of the elongated flexible circuit assembly and to the status indicators, wherein at least a portion of the plurality of visual status indicators are actuated, responsive to the control signal, to provide a visual indication of a status of equipment supported by the planar circuit board. In at least some embodiments, the multiplexing technique may include a time division multiplexing technique, in which some status indicators are activated, while others are deactivated in an alternative fashion and at a rate that may be imperceptible to a human observer. Alternatively or in addition, the computing environment **1200** can facilitate optimization processes that may identify any combination of logical groups of status indicators, status indicator actuation duty cycles, refresh rates, power utilization, power savings, and so on.

[0113] Generally, program modules comprise routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0114] As used herein, a processing circuit includes one or more processors as well as other application specific circuits such as an application specific integrated circuit, digital logic circuit, state machine, programmable gate array or other circuit that processes input signals or data and that produces output signals or data in response thereto. It should be noted that while any functions and

features described herein in association with the operation of a processor could likewise be performed by a processing circuit.

[0115] The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0116] Computing devices typically comprise a variety of media, which can comprise computer-readable storage media and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media can be any available storage media that can be accessed by the computer and comprises both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable instructions, program modules, structured data or unstructured data.

[0117] Computer-readable storage media can comprise, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0118] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0119] Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and comprises any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media comprise wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

[0120] With reference again to FIG. 12, the example environment can comprise a computer **1202**, the computer **1202** comprising a processing unit **1204**, a system memory **1206** and a system bus **1208**. The system bus **1208** couples system components including, but not limited to, the system memory **1206** to the processing unit **1204**. The processing unit **1204** can be any of various commercially available processors. Dual microprocessors and other multiprocessor architectures can also be employed as the processing unit **1204**.

[0121] The system bus **1208** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory **1206** comprises ROM **1210** and RAM **1212**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **1202**, such as during startup. The RAM **1212** can also comprise a high-speed RAM such as static RAM for caching data.

[0122] The computer **1202** further comprises an internal hard disk drive (HDD) **1214** (e.g., EIDE,

SATA), which internal HDD **1214** can also be configured for external use in a suitable chassis (not shown), a magnetic floppy disk drive (FDD) **1216**, (e.g., to read from or write to a removable diskette **1218**) and an optical disk drive **1220**, (e.g., reading a CD-ROM disk **1222** or, to read from or write to other high-capacity optical media such as the DVD). The HDD **1214**, magnetic FDD **1216** and optical disk drive **1220** can be connected to the system bus **1208** by a hard disk drive interface **1224**, a magnetic disk drive interface **1226** and an optical drive interface **1228**, respectively. The hard disk drive interface **1224** for external drive implementations comprises at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) **1394** interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

[0123] The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer **1202**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to a hard disk drive (HDD), a removable magnetic diskette, and a removable optical media such as a CD or DVD, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, such as zip drives, magnetic cassettes, flash memory cards, cartridges, and the like, can also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

[0124] A number of program modules can be stored in the drives and RAM **1212**, comprising an operating system **1230**, one or more application programs **1232**, other program modules **1234** and program data **1236**. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM **1212**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

[0125] A user can enter commands and information into the computer **1202** through one or more wired/wireless input devices, e.g., a keyboard **1238** and a pointing device, such as a mouse **1240**. Other input devices (not shown) can comprise a microphone, an infrared (IR) remote control, a joystick, a game pad, a stylus pen, touch screen or the like. These and other input devices are often connected to the processing unit **1204** through an input device interface **1242** that can be coupled to the system bus **1208**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a universal serial bus (USB) port, an IR interface, etc.

[0126] A monitor **1244** or other type of display device can be also connected to the system bus **1208** via an interface, such as a video adapter **1246**. It will also be appreciated that in alternative embodiments, a monitor **1244** can also be any display device (e.g., another computer having a display, a smart phone, a tablet computer, etc.) for receiving display information associated with computer **1202** via any communication means, including via the Internet and cloud-based networks. In addition to the monitor **1244**, a computer typically comprises other peripheral output devices (not shown), such as speakers, printers, etc.

[0127] The computer **1202** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **1248**. The remote computer(s) **1248** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically comprises many or all of the elements described relative to the computer **1202**, although, for purposes of brevity, only a remote memory/storage device **1250** is illustrated. The logical connections depicted comprise wired/wireless connectivity to a local area network (LAN) **1252** and/or larger networks, e.g., a wide area network (WAN) **1254**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

[0128] When used in a LAN networking environment, the computer **1202** can be connected to the

LAN **1252** through a wired and/or wireless communication network interface or adapter **1256**. The adapter **1256** can facilitate wired or wireless communication to the LAN **1252**, which can also comprise a wireless AP disposed thereon for communicating with the adapter **1256**.

[0129] When used in a WAN networking environment, the computer **1202** can comprise a modem **1258** or can be connected to a communications server on the WAN **1254** or has other means for establishing communications over the WAN **1254**, such as by way of the Internet. The modem **1258**, which can be internal or external and a wired or wireless device, can be connected to the system bus **1208** via the input device interface **1242**. In a networked environment, program modules depicted relative to the computer **1202** or portions thereof, can be stored in the remote memory/storage device **1250**. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

[0130] The computer **1202** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, restroom), and telephone. This can comprise Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0131] Wi-Fi can allow connection to the Internet from a couch at home, a bed in a hotel room or a conference room at work, without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks use radio technologies called IEEE 802.11 (a, b, g, n, ac, ag, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which can use IEEE 802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 5 GHz radio bands for example or with products that contain both bands (dual band), so the networks can provide real-world performance similar to the basic 10 BaseT wired Ethernet networks used in many offices.

[0132] The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is for clarity only and does not otherwise indicate or imply any order in time. For instance, “a first determination,” “a second determination,” and “a third determination,” does not indicate or imply that the first determination is to be made before the second determination, or vice versa, etc.

[0133] Moreover, it will be noted that the disclosed subject matter can be practiced with various computer system configurations, comprising single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as personal computers, hand-held computing devices (e.g., PDA, phone, smartphone, watch, tablet computers, netbook computers, etc.), microprocessor-based or programmable consumer or industrial electronics, and the like. The illustrated aspects can also be practiced in distributed computing environments where tasks are performed by remote processing devices that are linked through a communications network; however, some if not all aspects of the subject disclosure can be practiced on stand-alone computers. In a distributed computing environment, program modules can be in both local and remote memory storage devices.

[0134] Further, the various embodiments can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware, or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass semiconductor devices, wafers, integrated circuits, circuit modules, modules, systems and/or components incorporating semiconductor devices, as well as a computer program accessible from

any computer-readable device or computer-readable storage/communications media. For example, computer readable storage media can include, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various embodiments.

[0135] In addition, the words “example” and “exemplary” are used herein to mean serving as an instance or illustration. Any embodiment or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word example or exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

[0136] What has been described above includes mere examples of various embodiments. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing these examples, but one of ordinary skill in the art can recognize that many further combinations and permutations of the present embodiments are possible. Accordingly, the embodiments disclosed and/or claimed herein are intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

[0137] In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

[0138] As may also be used herein, the term(s) “operably coupled to,” “coupled to,” and/or “coupling” includes direct coupling between items and/or indirect coupling between items via one or more intervening items. Such items and intervening items include, but are not limited to, junctions, communication paths, components, circuit elements, circuits, functional blocks, and/or devices. As an example of indirect coupling, a signal conveyed from a first item to a second item may be modified by one or more intervening items by modifying the form, nature, or format of information in a signal, while one or more elements of the information in the signal are nevertheless conveyed in a manner than can be recognized by the second item. In a further example of indirect coupling, an action in a first item can cause a reaction on the second item, as a result of actions and/or reactions in one or more intervening items.

[0139] Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement which achieves the same or similar purpose may be substituted for the embodiments described or shown by the subject disclosure. The subject disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, can be used in the subject disclosure. For instance, one or more features from one or more embodiments can be

combined with one or more features of one or more other embodiments. In one or more embodiments, features that are positively recited can also be negatively recited and excluded from the embodiment with or without replacement by another structural and/or functional feature. The steps or functions described with respect to the embodiments of the subject disclosure can be performed in any order. The steps or functions described with respect to the embodiments of the subject disclosure can be performed alone or in combination with other steps or functions of the subject disclosure, as well as from other embodiments or from other steps that have not been described in the subject disclosure. Further, more than or less than all the features described with respect to an embodiment can also be utilized.

Claims

1. An elongated circuit assembly, comprising: a plurality of visual status indicators distributed along at least a distal end of the elongated circuit assembly; and a plurality of conductive paths that directs a group of control signals to the plurality of visual status indicators, wherein a number of the plurality of conductive paths is substantially fewer than a number of the plurality of visual status indicators.
2. The elongated circuit assembly of claim 1, wherein direction of the group of control signals is according to a multiplexing scheme.
3. The elongated circuit assembly of claim 2, wherein the multiplexing scheme allows a first portion of the plurality of conductive paths to provide a first portion of the group of control signals to a first portion of the plurality of visual status indicators, and wherein the multiplexing scheme allows a second portion of the plurality of conductive paths to provide a second portion of the group of control signals to a second portion of the plurality of visual status indicators.
4. The elongated circuit assembly of claim 2, wherein the multiplexing scheme comprises a time division multiplexing.
5. The elongated circuit assembly of claim 2, wherein the multiplexing scheme comprises spatial multiplexing.
6. The elongated circuit assembly of claim 1, wherein the number of the plurality of conductive paths is less than half the number of the plurality of visual status indicators.
7. The elongated circuit assembly of claim 1, wherein the plurality of visual status indicators comprises a group of light emitting diodes.
8. The elongated circuit assembly of claim 1, comprising a substrate.
9. The elongated circuit assembly of claim 8, wherein each of the plurality of visual status indicators is physically coupled to an outer surface of the substrate.
10. The elongated circuit assembly of claim 8, wherein the substrate comprises a group of substrate layers.
11. The elongated circuit assembly of claim 9, wherein the group of substrate layers comprises the plurality of conductive paths.
12. The elongated circuit assembly of claim 9, wherein a visual status indicator from the plurality of visual status indicators includes one or more electrical interconnections to the plurality of conductive paths within the group of substrate layers.
13. An electronic system, comprising: a plurality of visual status indicators; and a substrate, wherein each of the plurality of visual status indicators is physically coupled to an outer surface of the substrate, wherein the substrate comprises a group of substrate layers, wherein the group of substrate layers comprises a plurality of conductive paths that directs a group of control signals to the plurality of visual status indicators, wherein a number of the plurality of conductive paths is substantially fewer than a number of the plurality of visual status indicators.
14. The electronic system of claim 13, wherein a visual status indicator from the plurality of visual status indicators includes one or more electrical interconnections to the plurality of conductive

paths within the group of substrate layers.

15. The electronic system of claim 13, wherein the number of the plurality of conductive paths is less than half the number of the plurality of visual status indicators.

16. The electronic system of claim 13, wherein the plurality of visual status indicators comprises a group of light emitting diodes.

17. The electronic system of claim 13, wherein direction of the group of control signals is according to a multiplexing scheme.

18. The electronic system of claim 17, wherein the multiplexing scheme allows a first portion of the plurality of conductive paths to provide a first portion of the group of control signals to a first portion of the plurality of visual status indicators, and wherein the multiplexing scheme allows a second portion of the plurality of conductive paths to provide a second portion of the group of control signals to a second portion of the plurality of visual status indicators.

19. A method for providing status indication, comprising: receiving a group of control signals at a proximal end of an elongated circuit assembly; and directing the group of control signals along a plurality of conductive paths to a plurality of visual status indicators, wherein a number of the plurality of conductive paths is substantially fewer than a number of the plurality of visual status indicators.

20. The method of claim 19, wherein the directing of the group of control signals comprises multiplexing the group of control signals to allow a first portion of the plurality of conductive paths to provide a first portion of the group of control signals to a first portion of the plurality of visual status indicators and to allow allows a second portion of the plurality of conductive paths to provide a second portion of the group of control signals to a second portion of the plurality of visual status indicators.
